Kong ODE Benchmark

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<tr><th>Natural language statement</th><th>Domain</th><th>GroundTruth</th></tr>
<tr><td>Theorem 1.3.3. Let D be an open subset of R x Rn and (t0, a1, a2, ..., an) in D. (a) Assume g in C(D, Rn). Then there exists a > 0 such that IVP (1.3.10) has at least one solution which exists for |t - t0| <= . (b) Assume g in C(D, Rn), and as a function of (t, y1, y2, ..., yn), g is locally Lipschitz in (y1, y2, ..., yn) on D. Then there exists a > 0 such that IVP (1.3.10) has a unique solution which exists for |t - t0| <= .</td><td>ODE</td><td><pre>
def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
    (x : R -> (Fin n -> R)) : Prop :=
  forall t : R, HasDerivAt x (F t (x t)) t

def companionField {n : N} (g : R -> (Fin n -> R) -> R) (t : R)
    (y : (Fin n -> R)) : (Fin n -> R) :=
  fun i : Fin n => if h : i.1 + 1 < n then y <i.1 + 1, h> else g t y

def LocallyLipschitzInState {n : N} (D : Set (R x (Fin n -> R)))
    (f : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
  forall p in D, exists U in nhds p, exists K : NNReal,
    forall t : R, LipschitzOnWith K (f t) {x | (t, x) in U inter D}

theorem kong_1_3_3_nth_order_scalar_ivp
    {n : N} {D : Set (R x (Fin n -> R))} {g : R -> (Fin n -> R) -> R}
    {t0 : R} {a : (Fin n -> R)} (hD : IsOpen D) (hpoint : (t0, a) in D) :
    (ContinuousOn (fun p : R x (Fin n -> R) => g p.1 p.2) D ->
      exists : R, 0 < /\ let I := Set.Icc (t0 -) (t0 +)
        exists y : R -> (Fin n -> R), IsTrajectoryOn I (companionField g) y
        /\ y t0 = a /\
          forall t in I, (t, y t) in D) /\
    (ContinuousOn (fun p : R x (Fin n -> R) => g p.1 p.2) D ->
      LocallyLipschitzInState D (companionField g) ->
        exists : R, 0 < /\ let I := Set.Icc (t0 -) (t0 +)
          exists y : R -> (Fin n -> R), IsTrajectoryOn I (companionField g)
        y /\ y t0 = a /\
            (forall t in I, (t, y t) in D) /\
            forall z : R -> (Fin n -> R), IsTrajectoryOn I (companionField
        g) z ->
              z t0 = a -> (forall t in I, (t, z t) in D) -> Set.EqOn z y I)
        := by
  sorry
</pre></td></tr>
<tr><td>Theorem 1.5.3. Assume that f in C(D, Rn), df/dx in C(D, R^{n x n}), and df/dmu in C(D, R^{n x k}). Then IVP (V[t0, x0, mu]) has a unique solution x(t; t0, x0, mu) which is C1 in t0, x0, and mu in its domain. Furthermore, let J(t; t0, x0, mu) := df/dx(t, x(t; t0, x0, mu); mu). Then (a) (dx/dmu)(t; t0, x0, mu) is the solution of the IVP z’ = J(t; t0, x0, mu)z + df/dmu(t, x(t; t0, x0, mu); mu), z(t0)=0; (b) (dx/dx0)(t; t0, x0, mu) is the solution of z’ = J(t; t0, x0, mu)z, z(t0)=I; (c) (dx/dt0)(t; t0, x0, mu) is the solution of z’ = J(t; t0, x0, mu)z, z(t0)=-f(t0, x0; mu). Here I stands for the n x n identity matrix.</td><td>ODE</td><td><pre>
def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
    (x : R -> (Fin n -> R)) : Prop :=
  forall t : R, HasDerivAt x (F t (x t)) t

theorem kong_1_5_3_differentiable_dependence
    {n k : N} {D : Set (R x (Fin n -> R) x (Fin k -> R))}
    {f : R -> (Fin n -> R) -> (Fin k -> R) -> (Fin n -> R)}
    (hf : ContDiffOn R 1
      (fun p : R x (Fin n -> R) x (Fin k -> R) => f p.1 p.2.1 p.2.2) D)
      :
    exists x : R -> R -> (Fin n -> R) -> (Fin k -> R) -> (Fin n -> R),
      ((forall t0 x0 mu, (t0, x0, mu) in D ->
        IsTrajectory (fun t y => f t y mu) (fun t => x t t0 x0 mu) /\
          x t0 t0 x0 mu = x0) /\
        forall t0 x0 mu y, (t0, x0, mu) in D -> IsTrajectory (fun t z => f
        t z mu) y ->
          y t0 = x0 -> forall t, (t, y t, mu) in D -> y t = x t t0 x0 mu) /\
      ContDiff R 1 (fun p : R x R x (Fin n -> R) x (Fin k -> R) =>
        x p.1 p.2.1 p.2.2.1 p.2.2.2) /\
      (forall t0 x0 mu, let z := fun t => fderiv R (fun eta => x t t0 x0
      eta) mu
        z t0 = 0 /\ forall t, HasDerivAt z
          ((fderiv R (fun y => f t y mu) (x t t0 x0 mu)).comp (z t) +
            fderiv R (fun eta => f t (x t t0 x0 mu) eta) mu) t) /\
      (forall t0 x0 mu, let z := fun t => fderiv R (fun y => x t t0 y mu) x0
        z t0 = ContinuousLinearMap.id R (Fin n -> R) /\ forall t,
          HasDerivAt z ((fderiv R (fun y => f t y mu) (x t t0 x0 mu)).comp
        (z t)) t) /\
      forall t0 x0 mu, let z := fun t => deriv (fun s => x t s x0 mu) t0
        z t0 = -f t0 x0 mu /\ forall t,
          HasDerivAt z ((fderiv R (fun y => f t y mu) (x t t0 x0 mu)) (z
        t)) t := by
  sorry
</pre></td></tr>
<tr><td>Theorem 2.3.1 (Variation of Parameters Formula). Let X(t) be a fundamental matrix solution of Eq. (H) and t0 in (a,b). Then the general solution of Eq. (NH) is x = X(t)c + integral_{t0}^{t} X(t)X^{-1}(s)f(s) ds. In particular, the</td><td>ODE</td><td><pre>
def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
    (x : R -> (Fin n -> R)) : Prop :=
  forall t : R, HasDerivAt x (F t (x t)) t
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solution of the IVP consisting of Eq. (NH) and the IC x(t0)=x0 is x = X(t)X^{-1}(t0)x0 + integral_{t0}^{t} X(t)X^{-1}(s)f(s) ds.		`def FundamentalMatrixSolution {n : N} (I : Set R) (A : R -> Matrix (Fin n) (Fin n) R)`
		`    (X : R -> Matrix (Fin n) (Fin n) R) : Prop :=`
		`  (forall t in I, HasDerivAt X (A t * X t) t) /\ forall t in I, IsUnit (X t)`
		
		`theorem kong_2_3_1_variation_of_parameters`
		`    {n : N} {I : Set R} {A : R -> Matrix (Fin n) (Fin n) R} {f : R -> (Fin n -> R)}`
		`    {X : R -> Matrix (Fin n) (Fin n) R} {t0 : R}`
		`    (hX : FundamentalMatrixSolution I A X) (ht0 : t0 in I) :`
		`    (forall y : R -> (Fin n -> R),`
		`      (forall t in I, HasDerivAt y (A t *v y t + f t) t) <->`
		`        exists c : Fin n -> R, forall t in I,`
		`          y t = X t *v c + integral s in t0..t, (X t * (X s)^-1) *v f s) /\`
		`      forall x0, exists y : R -> (Fin n -> R), y t0 = x0 /\`
		`        (forall t in I, HasDerivAt y (A t *v y t + f t) t) /\`
		`        (forall t in I, y t = (X t * (X t0)^-1) *v x0 +`
		`          integral s in t0..t, (X t * (X s)^-1) *v f s) /\`
		`        forall z : R -> (Fin n -> R), z t0 = x0 ->`
		`          (forall t in I, HasDerivAt z (A t *v z t + f t) t) -> Set.EqOn z`
		`        y I := by`
		`  sorry`

Theorem 2.5.3 (Floquet Theorem). Let X(t) be a fundamental matrix solution of Eq. (H-p). Then there exists an R in C^{n x n} and a nonsingular omega-periodic P in C1(R, C^{n x n}) such that X(t) = P(t)e^{Rt}.	ODE	`def FundamentalMatrixSolution {n : N} (I : Set R) (A : R -> Matrix (Fin n) (Fin n) R)`
		`    (X : R -> Matrix (Fin n) (Fin n) R) : Prop :=`
		`  (forall t in I, HasDerivAt X (A t * X t) t) /\ forall t in I, IsUnit (X t)`
		
		`def PeriodicLinearEquation {n : N} (omega : R) (A : R -> Matrix (Fin n) (Fin n) R) : Prop :=`
		`  forall t, A (t + omega) = A t`
		
		`theorem kong_2_5_3_floquet_theorem`
		`    {n : N} {omega : R} {A : R -> Matrix (Fin n) (Fin n) R}`
		`    {X : R -> Matrix (Fin n) (Fin n) R}`
		`    (homega : 0 < omega) (hper : PeriodicLinearEquation omega A)`
		`    (hX : FundamentalMatrixSolution univ A X) :`
		`    exists R : Matrix (Fin n) (Fin n) C, exists P : R -> Matrix (Fin n) (Fin n) C,`
		`      (forall i j, ContDiff R 1 fun t => P t i j) /\ (forall t, P (t + omega) = P t) /\`
		`        (forall t, IsUnit (P t)) /\`
		`          forall t, (X t).map (algebraMap R C) = P t * NormedSpace.exp (t * R) := by`
		`  sorry`

Theorem 3.2.3. Let mui, i = 1, ..., n, be the characteristic multipliers of Eq. (H-p). Then (a) Equation (H-p) is uniformly stable => |mui| <= 1, i = 1, ..., n, and |mui| = 1 occurs only when the mui's are in the diagonal Jordan block of the transition matrix V; (b) Equation (H-p) is asymptotically stable => |mui| < 1, i = 1, ..., n; (c) Equation (H-p) is unstable => there exists an i in {1, ..., n} such that either |mui| > 1, or |mui| = 1 which occurs when mui is not in the diagonal Jordan block of the transition matrix V.	ODE	`def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))`
		`    (x : R -> (Fin n -> R)) : Prop :=`
		`  forall t : R, HasDerivAt x (F t (x t)) t`
		
		`def CharacteristicMultipliers {n : N} (V : Matrix (Fin n) (Fin n) C)`
		`    (mu : Fin n -> C) : Prop :=`
		`  (forall i, Matrix.det (mu i  *  (1 : Matrix (Fin n) (Fin n) C) - V) = 0) /\`
		`    forall z : C, Matrix.det (z  *  (1 : Matrix (Fin n) (Fin n) C) - V) = 0`
		`    -> exists i, mu i = z`
		
		`def InDiagonalJordanBlock {n : N} (V : Matrix (Fin n) (Fin n) C) (mu : C) : Prop :=`
		`  forall v : (Fin n -> C),`
		`    (V - mu  *  (1 : Matrix (Fin n) (Fin n) C)) *v`
		`        ((V - mu  *  (1 : Matrix (Fin n) (Fin n) C)) *v v) = 0 ->`
		`      (V - mu  *  (1 : Matrix (Fin n) (Fin n) C)) *v v = 0`
		
		`def UniformlyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop :=`
		`  UniformlyStableZeroSolution fun t x => A t *v x`
		
		`def AsymptoticallyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) : Prop :=`
		`  AsymptoticallyStableZeroSolution fun t x => A t *v x`

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		def UnstableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n) R) :
		Prop :=
		  UnstableZeroSolution fun t x => A t *v x
		
		theorem kong_3_2_3_characteristic_multiplier_stability
		    {n : N} {A : R -> Matrix (Fin n) (Fin n) R} {mu : Fin n -> C}
		    {V : Matrix (Fin n) (Fin n) C} {omega : R}
		    (homega : 0 < omega) (hperiodic : PeriodicLinearEquation omega A)
		    (hV : IsPeriodTransitionMatrix omega A V) (hmu :
		    CharacteristicMultipliers V mu) :
		    (UniformlyStableLinearEquation A <->
		      forall i, ||mu i|| <= 1 /\ (||mu i|| = 1 -> InDiagonalJordanBlock V
		      (mu i))) /\
		    (AsymptoticallyStableLinearEquation A <-> forall i, ||mu i|| < 1) /\
		    (UnstableLinearEquation A <->
		      exists i, 1 < ||mu i|| \/ (||mu i|| = 1 /\ not InDiagonalJordanBlock
		      V (mu i))) := by
		  sorry

Theorem 3.4.2. Assume that there exists a function p in C([0, inf), [0, inf)) such that integral_0^inf p(t) dt < inf, and |r(t, x)| <= p(t)|x| for sufficiently small |x| and all t in [0, inf). (a) If Eq. (H) is uniformly stable, then the zero solution of Eq. (3.4.6) is uniformly stable. (b) If Eq. (H) is uniformly stable and asymptotically stable, then the zero solution of Eq. (3.4.6) is uniformly stable and asymptotically stable.	ODE	def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
		    (x : R -> (Fin n -> R)) : Prop :=
		  forall t : R, HasDerivAt x (F t (x t)) t
		
		def IntegrableSmallPerturbation {n : N} (p : R -> R)
		    (r : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
		  ContinuousOn p (Set.Ici 0) /\ (forall t, 0 <= p t) /\
		    IntegrableOn p (Set.Ici 0) /\
		    exists rho : R, 0 < rho /\
		      forall t, 0 <= t -> forall x, ||x|| < rho -> ||r t x|| <= p t * ||x||
		
		def UniformlyStableLinearEquation {n : N} (A : R -> Matrix (Fin n) (Fin n)
		R) : Prop :=
		  UniformlyStableZeroSolution fun t x => A t *v x
		
		def AsymptoticallyStableLinearEquation {n : N} (A : R -> Matrix (Fin n)
		(Fin n) R) : Prop :=
		  AsymptoticallyStableZeroSolution fun t x => A t *v x
		
		def UniformlyStableZeroSolution {n : N}
		    (F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
		  forall eps : R, 0 < eps -> exists delta : R, 0 < delta /\ forall t0 x,
		    IsTrajectory F x -> ||x t0|| < delta -> forall t, t0 <= t -> ||x t|| <
		    eps
		
		def AsymptoticallyStableZeroSolution {n : N}
		    (F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=
		  UniformlyStableZeroSolution F /\ exists delta : R, 0 < delta /\ forall t0
		  x,
		    IsTrajectory F x -> ||x t0|| < delta -> Tendsto x atTop (nhds 0)
		
		theorem kong_3_4_2_integrable_perturbation_stability
		    {n : N} {A : R -> Matrix (Fin n) (Fin n) R}
		    {r : R -> (Fin n -> R) -> (Fin n -> R)} {p : R -> R}
		    (hr : IntegrableSmallPerturbation p r) :
		    (UniformlyStableLinearEquation A ->
		      UniformlyStableZeroSolution (fun t x => A t *v x + r t x)) /\
		    (UniformlyStableLinearEquation A -> AsymptoticallyStableLinearEquation
		    A ->
		      AsymptoticallyStableZeroSolution (fun t x => A t *v x + r t x)) := by
		  sorry

Theorem 3.5.2. Let D = {x in Rn : |x| <= l} for some l > 0 and V in C1(D, R). Assume that V(x) is positive definite and V(x) is negative semi-definite. Moreover, if the set D0 := {x in D : V(x) = 0} does not contain any nontrivial orbit of Eq. (A), then the zero solution of Eq. (A) is uniformly stable and asymptotically stable.	ODE	def IsTrajectory {n : N} (F : R -> (Fin n -> R) -> (Fin n -> R))
		    (x : R -> (Fin n -> R)) : Prop :=
		  forall t : R, HasDerivAt x (F t (x t)) t
		
		def LyapunovFunctionOnBall {n : N} (l : R) (V : (Fin n -> R) -> R)
		    (F : (Fin n -> R) -> (Fin n -> R)) : Prop :=
		  ContDiffOn R 1 V (Metric.closedBall 0 l) /\ V 0 = 0 /\
		    (forall x in Metric.closedBall (0 : (Fin n -> R)) l, x != 0 -> 0 < V x)
		    /\

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		`forall x in Metric.closedBall (0 : (Fin n -> R)) l, fderiv R V x (F x)`
		`<= 0`
		
		`def NoNontrivialOrbitInZeroDerivativeSet {n : N} (l : R)`
		`(V : (Fin n -> R) -> R) (F : (Fin n -> R) -> (Fin n -> R)) : Prop :=`
		`forall x : R -> (Fin n -> R), IsAutonomousTrajectory F x ->`
		`(forall t, x t in Metric.closedBall (0 : (Fin n -> R)) l /\`
		`fderiv R V (x t) (F (x t)) = 0) ->`
		`x = 0`
		
		`def UniformlyStableZeroSolution {n : N}`
		`(F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=`
		`forall eps : R, 0 < eps -> exists delta : R, 0 < delta /\ forall t0 x,`
		`IsTrajectory F x -> ||x t0|| < delta -> forall t, t0 <= t -> ||x t|| <`
		`eps`
		
		`def AsymptoticallyStableZeroSolution {n : N}`
		`(F : R -> (Fin n -> R) -> (Fin n -> R)) : Prop :=`
		`UniformlyStableZeroSolution F /\ exists delta : R, 0 < delta /\ forall t0`
		`x,`
		`IsTrajectory F x -> ||x t0|| < delta -> Tendsto x atTop (nhds 0)`
		
		`theorem kong_3_5_2_lasalle_invariance_stability`
		`{n : N} {l : R} {F : (Fin n -> R) -> (Fin n -> R)} {V : (Fin n -> R) ->`
		`R}`
		`(hl : 0 < l) (hV : LyapunovFunctionOnBall l V F)`
		`(horbit : NoNontrivialOrbitInZeroDerivativeSet l V F) :`
		`AsymptoticallyStableZeroSolution (fun _ x => F x) := by`
		`sorry`

Theorem 4.5.3. Let x(t) be a solution of system (A-2) and Gamma its orbit. Assume Gamma^+ is contained in a compact set E subset R2 and system (A-2) has at most a finite number of equilibria in E. Then one of the following four statements is true: (a) Omega(Gamma^+) contains only one equilibrium of system (A-2); (b) Gamma is a closed orbit; (c) Omega(Gamma^+) is a closed orbit; (d) Omega(Gamma^+) is a graphic for System (A-2). The same conclusion holds when Gamma^+ and Omega(Gamma^+) are replaced by Gamma^- and A(Gamma^-), respectively.	ODE	`def IsAutonomousTrajectory {n : N} (F : (Fin n -> R) -> (Fin n -> R))`
		`(x : R -> (Fin n -> R)) : Prop :=`
		`IsTrajectory (fun _ => F) x`
		
		`def omegaLimitSet {n : N} (x : R -> (Fin n -> R)) : Set (Fin n -> R) :=`
		`{y | exists t : N -> R, Tendsto t atTop atTop /\ Tendsto (fun j => x (t`
		`j)) atTop (nhds y)}`
		
		`def IsClosedOrbit {n : N} (F : (Fin n -> R) -> (Fin n -> R))`
		`(x : R -> (Fin n -> R)) : Prop :=`
		`IsAutonomousTrajectory F x /\ (exists t, x t != x 0) /\`
		`exists T : R, 0 < T /\ forall t, x (t + T) = x t`
		
		`def GraphicForPlanarSystem (F : (Fin 2 -> R) -> (Fin 2 -> R))`
		`(Gamma : Set (Fin 2 -> R)) : Prop :=`
		`IsConnected Gamma /\`
		`exists (m : N) (e : Fin m -> (Fin 2 -> R)) (x : Fin m -> R -> (Fin 2 ->`
		`R)),`
		`(forall i, F (e i) = 0) /\ (forall i, IsAutonomousTrajectory F (x i)) /\`
		`Gamma = range e union union i, range (x i)`
		
		`theorem kong_4_5_3_generalized_poincare_bendixson`
		`{F : (Fin 2 -> R) -> (Fin 2 -> R)} {x : R -> (Fin 2 -> R)}`
		`{E : Set (Fin 2 -> R)}`
		`(hcompact : IsCompact E) (horbit : IsAutonomousTrajectory F x /\ forall`
		`t, 0 <= t -> x t in E)`
		`(hfinite : {x in E | F x = 0}.Finite) :`
		`(exists e, omegaLimitSet x = {e} /\ F e = 0) \/ IsClosedOrbit F x \/`
		`(exists y, IsClosedOrbit F y /\ omegaLimitSet x = range y) \/`
		`GraphicForPlanarSystem F (omegaLimitSet x) := by`
		`sorry`

Theorem 5.4.2. Assume (5.4.2) and (5.4.4) hold with alpha'(0) = 0. Then either (a) all orbits of system (5.4.1-0) in a neighborhood of (0,0) are closed orbits and system (5.4.1-mu) does not have closed orbits for mu != 0 in a neighborhood of mu = 0, or (b) for mu > 0 sufficiently close to zero only or for mu < 0 sufficiently close to zero only, system (5.4.1-mu) has a unique limit cycle Gamma(mu) satisfying Gamma(mu) -> (0,0) with its period T(mu) -> 2pi/beta as mu -> 0.	ODE	`def IsAutonomousTrajectory {n : N} (F : (Fin n -> R) -> (Fin n -> R))`
		`(x : R -> (Fin n -> R)) : Prop :=`
		`IsTrajectory (fun _ => F) x`
		
		`def IsClosedOrbit {n : N} (F : (Fin n -> R) -> (Fin n -> R))`
		`(x : R -> (Fin n -> R)) : Prop :=`
		`IsAutonomousTrajectory F x /\ (exists t, x t != x 0) /\`
		`exists T : R, 0 < T /\ forall t, x (t + T) = x t`

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noncomputable def linearizationMatrix (F : (Fin 2 -> R) -> R -> (Fin 2 ->
R))
    (mu : R) : (Matrix (Fin 2) (Fin 2) R) :=
  fun i j => fderiv R (fun x => F x mu) 0 (Pi.single j 1) i

theorem kong_5_4_2_hopf_friedrich_dichotomy
    {F : (Fin 2 -> R) -> R -> (Fin 2 -> R)} {beta : R}
    (hbeta : 0 < beta)
    (hF : ContDiff R top (fun p : (Fin 2 -> R)  x  R => F p.1 p.2) /\
      (forall mu, F 0 mu = 0) /\ Matrix.trace (linearizationMatrix F 0) = 0
      /\
        Matrix.det (linearizationMatrix F 0) = beta ^ 2 /\
          HasDerivAt (fun mu => Matrix.trace (linearizationMatrix F mu)) 0
        0) :
    let center :=
      (exists eps : R, 0 < eps /\
        forall x : R -> (Fin 2 -> R), IsAutonomousTrajectory (fun y => F y
        0) x ->
          ||x 0|| < eps -> (exists t, x t != x 0) -> IsClosedOrbit (fun y
        => F y 0) x) /\
      exists eps : R, 0 < eps /\ forall mu, 0 < |mu| -> |mu| < eps ->
        not exists x : R -> (Fin 2 -> R), IsClosedOrbit (fun y => F y mu) x
        /\ ||x 0|| < eps
    let hopf := fun positiveSide =>
      exists eps : R, exists orbit : R -> R -> (Fin 2 -> R),
        exists period : R -> R, 0 < eps /\
        (forall mu, 0 < |mu| -> |mu| < eps -> (if positiveSide then 0 < mu
        else mu < 0) ->
          IsClosedOrbit (fun y => F y mu) (orbit mu) /\ 0 < period mu /\
            (forall t, orbit mu (t + period mu) = orbit mu t) /\
            forall y, IsClosedOrbit (fun z => F z mu) y -> ||y 0|| < eps ->
              range y = range (orbit mu)) /\
        Tendsto (fun mu => ||orbit mu 0||) (nhds[!=] 0) (nhds 0) /\
        Tendsto period (nhds[!=] 0) (nhds (2 * Real.pi / beta))
    center \/ hopf true \/ hopf false := by
  sorry
```

</td></tr>
<tr><td>Theorem 6.6.4. SLP (S-L), (P) has a countably infinite number
of eigenvalues lambdan, n in N0, which are all real, bounded
below and unbounded above, and can be arranged to satisfy the
coupling relations with mun and nun for n in N0: nu0 <=
lambda0 < {mu0,nu1} < lambda1 <= {mu1,nu2} <= lambda2 <
{mu2,nu3} < lambda3 <= {mu3,nu4} <= lambda4 < ... <
{mu_{2n},nu_{2n+1}} < lambda_{2n+1} <= {mu_{2n+1},nu_{2n+2}}
<= lambda_{2n+2} < .... Furthermore, (a) lambda0 is
geometrically simple; and for n >= 1, lambdan may be
geometrically simple or double, and lambdan is geometrically
double => lambdan = mui = nuj for some i,j in N0. (b) The
eigenfunctions associated with lambda0 have no zeros in [a,b],
and for n in N0, the eigenfunctions associated with
lambda_{2n+1} and lambda_{2n+2} have exactly 2n+2 zeros in the
half-open interval [a,b).</td><td>ODE</td><td>

```
def IsSturmLiouvilleEigenfunction (p q w : R -> R) (a b eigVal : R)
    (boundary : (R -> R) -> Prop) (y : R -> R) : Prop :=
  y != 0 /\ boundary y /\ exists y' : R -> R,
    (forall x in Set.Icc a b, HasDerivAt y (y' x) x) /\
    forall x in Set.Icc a b,
      HasDerivAt (fun t => p t * y' t) ((q x - eigVal * w x) * y x) x

def PeriodicSturmLiouvilleData (p q w : R -> R) (a b : R) : Prop :=
  a < b /\ ContinuousOn p (Set.Icc a b) /\ ContinuousOn q (Set.Icc a b) /\
    ContinuousOn w (Set.Icc a b) /\
    (forall x in Set.Icc a b, 0 < p x /\ 0 < w x)

def periodicBoundary (p : R -> R) (a b : R) (y : R -> R) : Prop :=
  y a = y b /\ deriv y a * p a = deriv y b * p b

def dirichletBoundary (a b : R) (y : R -> R) : Prop := y a = 0 /\ y b = 0

def neumannBoundary (a b : R) (y : R -> R) : Prop := deriv y a = 0 /\ deriv
y b = 0

theorem kong_6_6_4_periodic_sturm_liouville_coupling
    {p q w : R -> R} {a b : R} {lam mu nu : N -> R}
    (hSL : PeriodicSturmLiouvilleData p q w a b) :
    (forall n, lam n <= lam (n + 1)) /\ Tendsto lam atTop atTop /\
      (forall n, lam (2 * n + 1) <= mu n /\ mu n <= lam (2 * n + 2)) /\
      (forall n, lam (2 * n + 1) <= nu n /\ nu n <= lam (2 * n + 2)) /\
      (forall eigVal, (exists y, IsSturmLiouvilleEigenfunction p q w a b
      eigVal
        (periodicBoundary p a b) y) <-> exists n, lam n = eigVal) /\
      (forall eigVal, (exists y, IsSturmLiouvilleEigenfunction p q w a b
      eigVal
        (dirichletBoundary a b) y) <-> exists n, mu n = eigVal) /\
      (forall eigVal, (exists y, IsSturmLiouvilleEigenfunction p q w a b
      eigVal
```

</td></tr>
</table>

Kong ODE Benchmark

Natural language statement	Domain	GroundTruth

		` (neumannBoundary a b) y) <-> exists n, nu n = eigVal) /\`
		` (forall y, IsSturmLiouvilleEigenfunction p q w a b (lam 0)`
		` (periodicBoundary p a b) y -> {x in Set.Icc a b | y x = 0} = empty)`
		` /\`
		` (forall n, (exists i j, lam n = mu i /\ lam n = nu j) <->`
		` exists y1 y2, IsSturmLiouvilleEigenfunction p q w a b (lam n)`
		` (periodicBoundary p a b) y1 /\`
		` IsSturmLiouvilleEigenfunction p q w a b (lam n)`
		` (periodicBoundary p a b) y2 /\ not exists c : R, y2 = c * y1)`
		` /\`
		` forall n y, IsSturmLiouvilleEigenfunction p q w a b (lam (2 * n + 1))`
		` (periodicBoundary p a b) y ->`
		` {x in Set.Ico a b | y x = 0}.ncard = 2 * n + 2 := by`
		`sorry`